

Field Theory Exam Solutions (Part I)

Semester Exam Responses

Question 1

(a) A cube of unit volume is placed in a uniform electric field with its edges parallel to the coordinate axes. Determine the total charge enclosed by the unit cube.

Ans: According to Gauss's Law, the total electric flux leaving a closed surface is equal to the total enclosed charge Q_{enc} divided by the permittivity. In differential form:

$$\nabla \cdot \vec{D} = \rho_v$$

Since the electric field $\vec{E}$ is uniform throughout the region, the electric flux density $\vec{D} = \epsilon \vec{E}$ is also a constant vector. The divergence of a constant vector field is zero:

$$\nabla \cdot \vec{D} = 0 \implies \rho_v = 0$$

Integrating over the volume of the unit cube:

$$Q_{\text{enc}} = \int_V \rho_v dv = 0$$

Conclusion: The total charge enclosed by the unit cube is **0 C**.

(b) Consider a ring charge of radius 10cm and uniform charge density +0.2 nC/m and also a disc charge of radius 20cm and uniform charge density σ nC/m² ... find the value of σ . Relative permittivity of the medium is 4.

Ans: Let the center of both the ring and disc be at the origin (0,0,0) in the $x-y$ plane. The point of interest is on the z -axis at height $z = 0.3$ m. Permittivity $\epsilon = \epsilon_0 \epsilon_r = 4\epsilon_0$.

Electric Field due to the Ring Charge at point on z -axis:

$$\vec{E}_{\text{ring}} = \frac{\lambda \cdot z \cdot R_1}{2\epsilon_0 \epsilon_r (R_1^2 + z^2)^{3/2}} \hat{k}$$

Given: $\lambda = 0.2 \times 10^{-9}$ C/m, $R_1 = 0.1$ m, $z = 0.3$ m.

$$|\vec{E}_{\text{ring}}| = \frac{(0.2 \times 10^{-9})(0.3)(0.1)}{2 \cdot \epsilon_0 \cdot 4 \cdot (0.1^2 + 0.3^2)^{3/2}} = \frac{0.006 \times 10^{-9}}{8\epsilon_0(0.1)^{3/2}} = \frac{0.006 \times 10^{-9}}{8\epsilon_0(0.03162)} = \frac{23.7 \times 10^{-12}}{\epsilon_0} \text{ V/m}$$

Electric Field due to the Disc Charge at point on z -axis:

$$\vec{E}_{\text{disc}} = \frac{\sigma_{\text{true}}}{2\epsilon_0 \epsilon_r} \left(1 - \frac{z}{\sqrt{R_2^2 + z^2}} \right) \hat{k}$$

Given: $R_2 = 0.2$ m, $\sigma_{\text{true}} = \sigma \times 10^{-9}$ C/m².

$$|\vec{E}_{\text{disc}}| = \frac{\sigma \times 10^{-9}}{8\epsilon_0} \left(1 - \frac{0.3}{\sqrt{0.2^2 + 0.3^2}} \right) = \frac{\sigma \times 10^{-9}}{8\epsilon_0} \left(1 - \frac{0.3}{\sqrt{0.13}} \right) = \frac{\sigma \times 10^{-9}}{8\epsilon_0} (1 - 0.832) = \frac{0.021\sigma \times 10^{-9}}{\epsilon_0} \text{ V/m}$$

Since the net electric field intensity at $z = 0.3$ m is zero:

$$\vec{E}_{\text{ring}} + \vec{E}_{\text{disc}} = 0 \implies |\vec{E}_{\text{ring}}| = -|\vec{E}_{\text{disc}}|$$

$$\frac{23.7 \times 10^{-12}}{\epsilon_0} + \frac{21\sigma \times 10^{-12}}{\epsilon_0} = 0 \implies 21\sigma = -23.7 \implies \sigma = -1.13$$

Conclusion: $\sigma = -1.13$ nC/m².

(c) Prove that $\nabla \cdot \vec{D} = \rho_v$.

Ans: From Gauss's Law in integral form, the total electric flux leaving a closed surface S bounds a volume V is equal to the total charge enclosed:

$$\oint_S \vec{D} \cdot d\vec{s} = Q_{\text{enc}} = \int_V \rho_v dv$$

By applying the Divergence Theorem, the closed surface integral of a vector field can be converted to a volume integral:

$$\oint_S \vec{D} \cdot d\vec{s} = \int_V (\nabla \cdot \vec{D}) dv$$

Equating the two expressions:

$$\int_V (\nabla \cdot \vec{D}) dv = \int_V \rho_v dv$$

Since this holds true for any arbitrary volume V , the integrands must be equal:

$$\nabla \cdot \vec{D} = \rho_v \quad (\text{Hence Proved})$$

Question 2

(a) Starting from the expression of mechanical pressure on a conductor surface, derive the expression for volume density of energy in an electric field.

Ans: The mechanical pressure P on a conductor surface in an electric field E is given by:

$$P = \frac{1}{2} \epsilon E^2 \text{ N/m}^2$$

Consider a small area dA on the conductor surface. The outward force acting on this area is:

$$dF = P dA = \frac{1}{2} \epsilon E^2 dA$$

If this elemental area moves outwards by a small distance dx under the influence of this force, the mechanical work done is:

$$dW = dF \cdot dx = \left(\frac{1}{2} \epsilon E^2 dA \right) dx$$

Notice that $dA \cdot dx = dV$, which is the small volume swept by the surface. Thus:

$$dW = \frac{1}{2} \epsilon E^2 dV$$

This work done is stored as electrostatic potential energy dU in the newly created field volume. Therefore, the energy stored per unit volume (Energy Density, w_e) is:

$$w_e = \frac{dU}{dV} = \frac{dW}{dV} = \frac{1}{2} \epsilon E^2 \text{ J/m}^3 \quad (\text{Derived})$$

(b) Point charge of +1 mC at (0,0,1) m, infinite conducting plane at z=0. Find V and electric stresses at (3,4,1) m.

Ans: Using the Method of Images, replace the conducting plane at $z = 0$ with an image charge $-Q = -1$ mC at $(0, 0, -1)$ m. Medium is air ($\epsilon_r = 1$). Let $Q = 10^{-3}$ C. Point of interest $P(3, 4, 1)$. Distance to Q at $(0, 0, 1)$: $R_1 = \sqrt{3^2 + 4^2 + (1-1)^2} = \sqrt{25} = 5$ m. Distance to $-Q$ at $(0, 0, -1)$: $R_2 = \sqrt{3^2 + 4^2 + (1-(-1))^2} = \sqrt{9 + 16 + 4} = \sqrt{29}$ m.

1. *Electric Potential (V):*

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{R_1} + \frac{-Q}{R_2} \right) = 9 \times 10^9 \times 10^{-3} \left(\frac{1}{5} - \frac{1}{\sqrt{29}} \right)$$

$$V = 9 \times 10^6 (0.2 - 0.1857) = 9 \times 10^6 (0.0143) = 1.287 \times 10^5 \text{ V} = \mathbf{128.7 \text{ kV}}$$

2. *Electric Stresses:* Stress is defined as $w_e = \frac{1}{2}\epsilon_0 |\vec{E}_{\text{net}}|^2$.

$$\vec{E}_1 = \frac{Q}{4\pi\epsilon_0 R_1^3} \vec{R}_1 = 9 \times 10^6 \frac{3\hat{i} + 4\hat{j}}{125} = 216\hat{i} + 288\hat{j} \text{ (kV/m)}$$

$$\vec{E}_2 = \frac{-Q}{4\pi\epsilon_0 R_2^3} \vec{R}_2 = -9 \times 10^6 \frac{3\hat{i} + 4\hat{j} + 2\hat{k}}{29\sqrt{29}} = -172.7\hat{i} - 230.2\hat{j} - 115.1\hat{k} \text{ (kV/m)}$$

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2 = 43.3\hat{i} + 57.8\hat{j} - 115.1\hat{k} \text{ (kV/m)}$$

$$|\vec{E}_{\text{net}}|^2 = (43.3)^2 + (57.8)^2 + (-115.1)^2 = 1874.89 + 3340.84 + 13248.01 = 18463.74 \text{ (kV/m)}^2 = 1.846 \times 10^{10} \text{ (V/m)}^2$$

$$\text{Stress} = 0.5 \times (8.854 \times 10^{-12}) \times (1.846 \times 10^{10}) = \mathbf{0.0817 \text{ N/m}^2}.$$

(c) Define "Electric Field Intensity" at a point.

Ans: Electric Field Intensity ($\vec{E}$) at any point is defined as the force experienced by a unit positive test charge placed at that point, assuming the test charge is small enough not to disturb the original source field. Mathematically:

$$\vec{E} = \lim_{q \rightarrow 0} \frac{\vec{F}}{q} \text{ (V/m or N/C)}$$

Question 3

(a) Establish boundary conditions for normal component of electric flux density at dielectric-dielectric interface.

Ans: Consider a boundary separating two perfect dielectrics (permittivities ϵ_1 and ϵ_2). Construct a small cylindrical Gaussian pillbox of height $\Delta h \rightarrow 0$ and cross-sectional area ΔS placed such that it is halved by the boundary. Applying Gauss's Law to the pillbox:

$$\oint \vec{D} \cdot d\vec{s} = Q_{\text{free_enclosed}}$$

Since $\Delta h \rightarrow 0$, the flux contribution from the curved side surface goes to zero. Flux only passes through the top and bottom flat surfaces.

$$\vec{D}_1 \cdot \Delta \vec{S} + \vec{D}_2 \cdot (-\Delta \vec{S}) = \rho_s \Delta S$$

where $\vec{D}_1$ and $\vec{D}_2$ are flux densities in medium 1 and 2, and ρ_s is the free surface charge density on the boundary. Taking normal components:

$$D_{1n} \Delta S - D_{2n} \Delta S = \rho_s \Delta S \implies D_{1n} - D_{2n} = \rho_s$$

For ideal dielectric-dielectric interfaces, there are no free charges, so $\rho_s = 0$. Hence:

$$D_{1n} = D_{2n} \implies \epsilon_1 E_{1n} = \epsilon_2 E_{2n}$$

Conclusion: The normal component of the electric flux density $\vec{D}$ is continuous across a charge-free dielectric boundary.

(b) Single-core cable, 3 different dielectric media. Explain condition for max E-field to NOT occur just off the inner conductor.

Ans: In a coaxial cable with inner radius r and outer sheath radius R , the electric field intensity at a distance x from the center is given by:

$$E_x = \frac{q}{2\pi\epsilon x}$$

For a graded cable with 3 dielectrics having permittivities $\epsilon_1, \epsilon_2, \epsilon_3$ and outer radii r_1, r_2, R respectively (where the innermost layer touches the conductor of radius r), the maximum electric field in each layer occurs at its innermost radius:

$$E_{1\max} = \frac{q}{2\pi\epsilon_1 r} \quad , \quad E_{2\max} = \frac{q}{2\pi\epsilon_2 r_1} \quad , \quad E_{3\max} = \frac{q}{2\pi\epsilon_3 r_2}$$

Normally, without grading, the maximum overall stress occurs at $x = r$ (just off inner conductor). To ensure that the global maximum stress does **not** occur just off the inner conductor, the maximum stress in a subsequent layer must exceed the maximum stress in the first layer. For example, if we require $E_{2\max} > E_{1\max}$:

$$\frac{q}{2\pi\epsilon_2 r_1} > \frac{q}{2\pi\epsilon_1 r} \implies \frac{1}{\epsilon_2 r_1} > \frac{1}{\epsilon_1 r} \implies \epsilon_1 r > \epsilon_2 r_1$$

Similarly, for the third layer to be the maximum: $\epsilon_1 r > \epsilon_3 r_2$. *Conclusion:* Improper grading where the product $\epsilon_{\text{inner}} \cdot r_{\text{inner}} > \epsilon_{\text{outer}} \cdot r_{\text{outer_inner_boundary}}$ forces the maximum field out of the innermost layer.

(c) Determine whether $\vec{E} = y\hat{i} + x\hat{j} + z\hat{k}$ is a valid form of electric field.

Ans: A valid static electric field must be conservative, meaning it must be irrotational. Its curl must be zero ($\nabla \times \vec{E} = 0$).

$$\begin{aligned} \nabla \times \vec{E} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & x & z \end{vmatrix} \\ &= \hat{i} \left(\frac{\partial}{\partial y}(z) - \frac{\partial}{\partial z}(x) \right) - \hat{j} \left(\frac{\partial}{\partial x}(z) - \frac{\partial}{\partial z}(y) \right) + \hat{k} \left(\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(y) \right) \\ &= \hat{i}(0 - 0) - \hat{j}(0 - 0) + \hat{k}(1 - 1) = \vec{0} \end{aligned}$$

Since $\nabla \times \vec{E} = 0$, the given field **is a valid** form of an electrostatic field.

Question 4

(a) Derive FDM equations for unknown node potentials in 2-D system with unequal nodal distances.

Ans: Consider a node 0 at coordinates (x, y) with potential V_0 . The neighboring nodes 1, 2, 3, 4

are at distances h_1, h_2, h_3, h_4 along $+x, +y, -x, -y$ directions respectively. Using Taylor series expansions for potential V along the x-axis:

$$V_1 = V_0 + h_1 \frac{\partial V}{\partial x} + \frac{h_1^2}{2!} \frac{\partial^2 V}{\partial x^2} + \dots$$

$$V_3 = V_0 - h_3 \frac{\partial V}{\partial x} + \frac{h_3^2}{2!} \frac{\partial^2 V}{\partial x^2} - \dots$$

Multiplying V_1 by h_3 and V_3 by h_1 , then adding them eliminates the first derivative:

$$h_3 V_1 + h_1 V_3 \approx (h_3 + h_1) V_0 + \frac{h_1 h_3 (h_1 + h_3)}{2} \frac{\partial^2 V}{\partial x^2}$$

$$\Rightarrow \frac{\partial^2 V}{\partial x^2} = \frac{2}{h_1 h_3 (h_1 + h_3)} [h_3 V_1 + h_1 V_3 - (h_1 + h_3) V_0] = \frac{2}{h_1 + h_3} \left[\frac{V_1 - V_0}{h_1} + \frac{V_3 - V_0}{h_3} \right]$$

Similarly, along the y-axis:

$$\frac{\partial^2 V}{\partial y^2} = \frac{2}{h_2 + h_4} \left[\frac{V_2 - V_0}{h_2} + \frac{V_4 - V_0}{h_4} \right]$$

For a charge-free region, Laplace's Equation applies ($\nabla^2 V = 0$):

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

$$\frac{2}{h_1 + h_3} \left[\frac{V_1 - V_0}{h_1} + \frac{V_3 - V_0}{h_3} \right] + \frac{2}{h_2 + h_4} \left[\frac{V_2 - V_0}{h_2} + \frac{V_4 - V_0}{h_4} \right] = 0$$

This is the required FDM difference equation for unequal nodal spacing.

(b) Calculate the capacitance between two infinitely long transmission line conductors.

Ans: Consider two parallel infinite conductors A and B of radius r , separated by a distance D (where $D \gg r$). Let conductor A carry linear charge density $+\lambda$ C/m and B carry $-\lambda$ C/m. Electric field at a distance x from center of A on the line joining their centers:

$$\vec{E} = \frac{\lambda}{2\pi\epsilon x} \hat{a}_x + \frac{\lambda}{2\pi\epsilon(D-x)} \hat{a}_x$$

Potential difference V_{AB} is the work done moving a unit positive charge from B to A:

$$V_{AB} = - \int_{D-r}^r E dx = \int_r^{D-r} \left(\frac{\lambda}{2\pi\epsilon x} + \frac{\lambda}{2\pi\epsilon(D-x)} \right) dx$$

$$V_{AB} = \frac{\lambda}{2\pi\epsilon} [\ln x - \ln(D-x)]_r^{D-r} = \frac{\lambda}{2\pi\epsilon} \left(\ln \frac{D-r}{r} - \ln \frac{r}{D-r} \right) = \frac{\lambda}{\pi\epsilon} \ln \left(\frac{D-r}{r} \right)$$

Since $D \gg r$, $D-r \approx D$, so $V_{AB} \approx \frac{\lambda}{\pi\epsilon} \ln(D/r)$. Capacitance per unit length $C = \frac{\lambda}{V_{AB}}$:

$$C = \frac{\pi\epsilon}{\ln(D/r)} \text{ F/m}$$

(c) Justify/Correct: "Electric field intensity just off the conductor surface is exactly equal to the electric field intensity on the conductor surface".

Ans: Incorrect. The correct statement is: Electric field intensity just off the conductor surface is **double** the field intensity exactly **on** the conductor surface. *Justification:* Using Gauss's Law,

the field *just off* the surface is $E = \sigma/\epsilon$. To find the field exactly *on* the surface, we divide the charge into two parts: a tiny local circular patch of charge, and the rest of the conductor's charge. The field just off the surface is the sum of field from the patch ($\sigma/2\epsilon$) and field from the rest ($\sigma/2\epsilon$), giving σ/ϵ . However, right *on* the conductor layer itself, the local patch produces no net field in the normal direction on itself. The force experienced by the charge is only due to the "rest of the charge", which provides a field of $E_{\text{on_surface}} = \sigma/2\epsilon$. Thus, $E_{\text{on_surface}} = \frac{1}{2}E_{\text{just_off}}$.

Question 5

(a) State and prove Uniqueness Theorem.

Ans: Statement: The Uniqueness Theorem states that if a solution to Laplace's or Poisson's equation satisfies the given boundary conditions for a specified region, then it is the *only* (unique) solution for that region.

Proof: Assume there are two different solutions, V_1 and V_2 , satisfying Laplace's equation ($\nabla^2 V_1 = 0$ and $\nabla^2 V_2 = 0$) in volume τ bounded by surface S . Since both satisfy the same boundary conditions, on boundary surface S , $V_1 = V_2$. Let us define a difference potential: $V_d = V_1 - V_2$. 1. $\nabla^2 V_d = \nabla^2(V_1 - V_2) = 0$ inside τ . 2. On boundary S , $V_d = V_1 - V_2 = 0$.

Applying the vector identity: $\nabla \cdot (V_d \nabla V_d) = V_d \nabla^2 V_d + |\nabla V_d|^2$. Since $\nabla^2 V_d = 0$, this reduces to: $\nabla \cdot (V_d \nabla V_d) = |\nabla V_d|^2$. Integrate over the volume τ and apply Divergence Theorem:

$$\int_{\tau} \nabla \cdot (V_d \nabla V_d) d\tau = \oint_S (V_d \nabla V_d) \cdot d\vec{s} = \int_{\tau} |\nabla V_d|^2 d\tau$$

Since $V_d = 0$ on the boundary surface S , the closed surface integral evaluates to 0.

$$\int_{\tau} |\nabla V_d|^2 d\tau = 0$$

Because $|\nabla V_d|^2$ is always strictly non-negative, its volume integral can only be zero if the integrand itself is identically zero everywhere in the volume.

$$|\nabla V_d|^2 = 0 \implies \nabla V_d = 0$$

If the gradient of a scalar is zero everywhere, the scalar must be a constant. $V_d = \text{constant}$. But since $V_d = 0$ on the boundary, the constant must be zero everywhere.

$$V_d = 0 \implies V_1 - V_2 = 0 \implies V_1 = V_2$$

Thus, there is only one unique solution. (Q.E.D)

(b) Prove that the equipotentials due to an infinitely long line charge and its image wrt an infinitely long conducting plane are cylinders...

Ans: Consider an infinitely long line charge $+\lambda$ located at $x = a$ and its image $-\lambda$ located at $x = -a$. The potential V at any point $P(x, y)$ due to these two line charges is given by the superposition of their individual potentials:

$$V_P = -\frac{\lambda}{2\pi\epsilon_0} \ln R_1 + \frac{\lambda}{2\pi\epsilon_0} \ln R_2 = \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{R_2}{R_1} \right)$$

Where R_1 is the distance from $+\lambda$ to P , and R_2 is the distance from $-\lambda$ to P .

$$R_1 = \sqrt{(x-a)^2 + y^2} \quad , \quad R_2 = \sqrt{(x+a)^2 + y^2}$$

For an equipotential surface, $V_P = \text{constant} = V_0$. Therefore:

$$\frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{R_2}{R_1}\right) = V_0 \implies \frac{R_2}{R_1} = e^{2\pi\epsilon_0 V_0/\lambda} = K \text{ (a constant)}$$

Squaring both sides:

$$\frac{(x+a)^2 + y^2}{(x-a)^2 + y^2} = K^2$$

$$(x+a)^2 + y^2 = K^2[(x-a)^2 + y^2]$$

Expanding and gathering terms:

$$x^2 + 2ax + a^2 + y^2 = K^2(x^2 - 2ax + a^2 + y^2)$$

$$x^2(K^2 - 1) - 2ax(K^2 + 1) + y^2(K^2 - 1) + a^2(K^2 - 1) = 0$$

Dividing by $(K^2 - 1)$:

$$x^2 - 2x\left[a\frac{K^2 + 1}{K^2 - 1}\right] + y^2 + a^2 = 0$$

This can be written by completing the square for x :

$$\left(x - a\frac{K^2 + 1}{K^2 - 1}\right)^2 - \left(a\frac{K^2 + 1}{K^2 - 1}\right)^2 + y^2 + a^2 = 0$$

$$\left(x - a\frac{K^2 + 1}{K^2 - 1}\right)^2 + y^2 = a^2 \left[\left(\frac{K^2 + 1}{K^2 - 1}\right)^2 - 1\right]$$

This represents the standard equation of a circle $(x-h)^2 + y^2 = R^2$ in the $x-y$ plane. Since the system is infinite along the z -axis, this 2D circular cross-section sweeps out a cylinder in 3D space. Thus, the equipotential surfaces are cylinders with axes parallel to the line charges. (Hence Proved)